

Sabarinath B

Student

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Profile

I'm a second-year B.Tech student in Artificial Intelligence and Data Science at Amrita Vishwa Vidyapeetham. I enjoy building systems from scratch, ranging from software applications to IoT-based sensing projects. My work includes simulation studies, backend development, and research on computer-vision-based wildlife monitoring using infrared-triggered detection and IoT alerting, which was presented at IEEE SCEECS'26. I'm currently focused on improving my programming and machine learning skills through hands-on projects and research.

Education

B.Tech – Artificial Intelligence and Data Science 2024 – Present
Amrita Vishwa Vidyapeetham, Amritapuri Campus, Kerala, India
CGPA: 8.17

Higher Senior Secondary (+2) – CBSE 2021 – 2023
Sree Narayana Public School, Kollam
Score: 91%
Courses: PCM + Computer Science

Projects

- **IoT-Based Wild Animal Detection and Alert System (2024)**
Developed an IoT-based system using infrared sensors to detect wild animals and trigger real-time alerts to nearby residents and authorities.
- **Enhancing Predictive Maintenance in Military Robotic Systems – GRU Based Failure Prediction (2025)**
Designed a deep learning framework for predictive maintenance in robotic systems using an Attention-based GRU model to capture temporal degradation patterns and perform multi-class failure prediction.
(Submitted to IEEE Access — currently under review.)
- **Transient Analysis of Piston, Crankshaft and Connecting Rod – ANSYS (2025)**
Performed transient structural analysis using ANSYS, including dynamic load simulation, mesh optimization, stress analysis, and validation of structural integrity.
- **Goods Vehicle Booking & Management System – Java (2025)**
Developed a Java-based vehicle booking and management system using MySQL and AWT/Swing for managing drivers, customers, and trip scheduling.
- **Tryno 2.0 – Discord Bot (2018)**
Built an interactive Discord chatbot with API-based command handling and automation features.
- **Character Design – Blender (2022)**
Created a photorealistic 3D character model using Blender, focusing on detailed modeling, texturing, and rendering.

Experience

Research, Fieldwork, and Freelance

Live-in-Labs[®] Volunteer Ongoing
Community-based research and fieldwork for Chidambarandhapuram village
Live-in-Labs[®] Program

Freelancing – Design & Digital Services

2021 – 2022

- Poster design for gaming tournament announcements — Vellore Institute of Technology (VIT Thanima), Vellore, Tamil Nadu
- Website and logo design for a construction firm — Manal Constructions
- Digital marketing support— AZ Digital Marketing Agency, Canada

Publications & Research

A Hybrid Framework for Human–Wildlife Conflict Mitigation Using Infrared-Triggered YOLO and IoT Alerting

Presented at IEEE SCEECs'26 Conference, IEEE MANIT Student Branch

January 31 – February 1, 2026

Paper ID: 474

Publication pending in IEEE Xplore

Skills

Programming Languages: Python, Java, JavaScript, R

Web Development: HTML, CSS, React.js

Databases: MySQL, MongoDB

Tools & Software: Git, MATLAB, ANSYS, Blender, Photoshop, GIMP

Video Editing: Premiere Pro, DaVinci Resolve

Game Development: Unreal Engine, Godot

Soft Skills: Structured Thinking, Creativity

Certifications

NPTEL – Data Science for Engineers

Certificate ID: NPTEL25CS101S539700157

MATLAB Onramp

MathWorks Academy

Deep Learning in Python

DataCamp (18 hours, February 2026)

Soft Skills Training

Infosys Springboard, Course Completion Certificate (October 2025)

Workshops

Artificial Intelligence and Machine Learning Applications – NIT Kurukshetra (08/2024)

Artificial Intelligence and Machine Learning with Python – NIT Calicut (09/2024)

Data Mining – NIT Calicut (09/2024)

Hacktoberfest – amFOSS (10/2024)

Languages

English — Malayalam (native) — Hindi

Interests

Programming, 3D Modelling, PC Gaming, Numismatics, Reading, Technology